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APPROACH TO ANEMIA IN CHILDREN AND NUTRITIONAL ANEMIA

WHO Definition :

Age group	Hemoglobin (g/dL) cut-off
6 months - 5 years	<11
5-11 years	<11.5
12-13 years	<12
Non-pregnant women	<12
men	<13

Note : Hb is more in m>f d/t effect of androgens.

Nomograms :

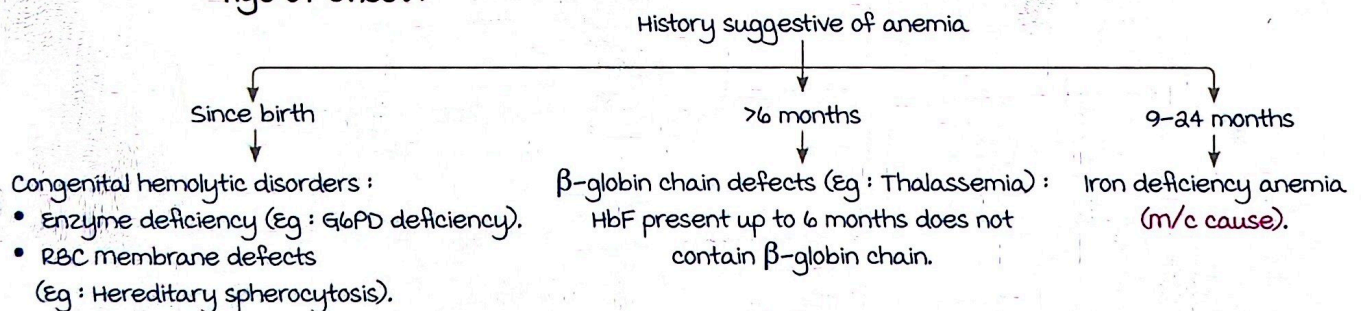
- **Best method** to define anemia in a child (Different graphs for males & females).
- Based on Hemoglobin & Hematocrit.
- Anemia : < **2SD from mean** for age & gender (<-2SD).

Approach to Anemia

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HISTORY :

Age of Onset :



Diet History :

- Strict vegetarian/vegan : vit B₁₂ deficiency.
 - Goat's milk diet : Folic acid deficiency.
 - Exclusive Cow milk diet : Iron & vit C deficiency (vit C aids iron absorption).
 - Pica (Eating inedible substances) : Iron deficiency anemia.
- **megaloblastic anemia**

Other History : khanirfan0392@gmail.com

- H/o chronic disease : Anemia of chronic inflammation.
- Family History : **Hemolytic anemia** → - Anemia + jaundice.
- Gallstone.
- Splenectomy.

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Page 1/6

FOCUSED PHYSICAL EXAMINATION

Universal feature : Pallor.

Associated Congenital Defects :

- Skeletal defects : Absent radius/thumb.
- Seen in **Fanconi anemia** (Bone marrow failure).

Splenomegaly :

- Hemolytic anemia.
- A/w hepatomegaly : Leukemia.

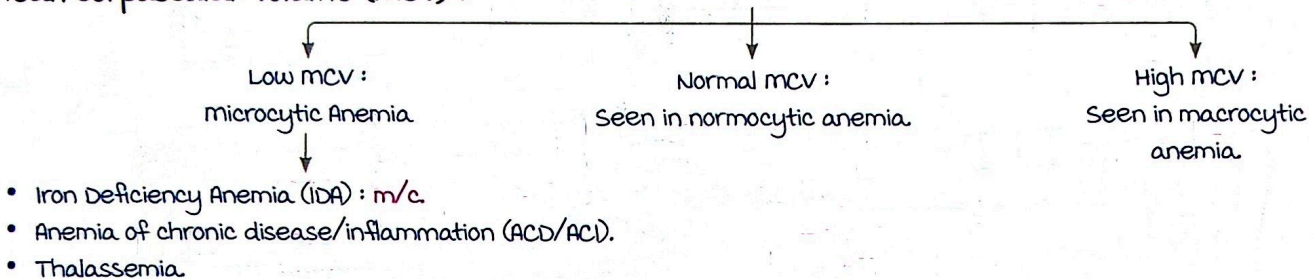
Petechiae/purpura (Thrombocytopenia) :

- Leukemia.
- Aplastic anemia.
- Hemolytic uremic syndrome (Acute onset).

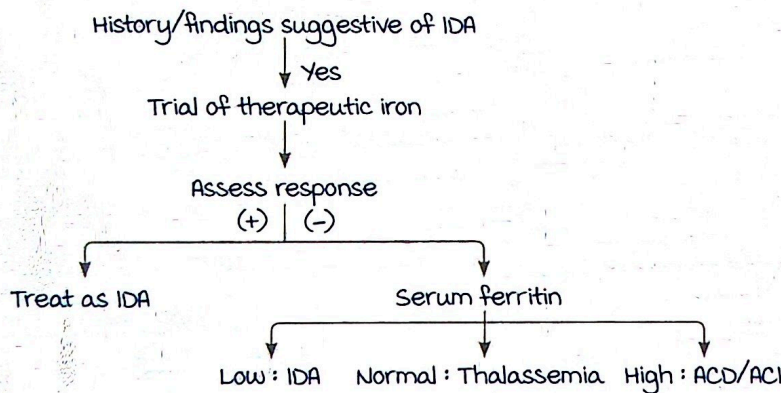
Generalized lymphadenopathy : Leukemia.

INVESTIGATIONS :

mean corpuscular volume (mcv) :



Low MCV Anemia :

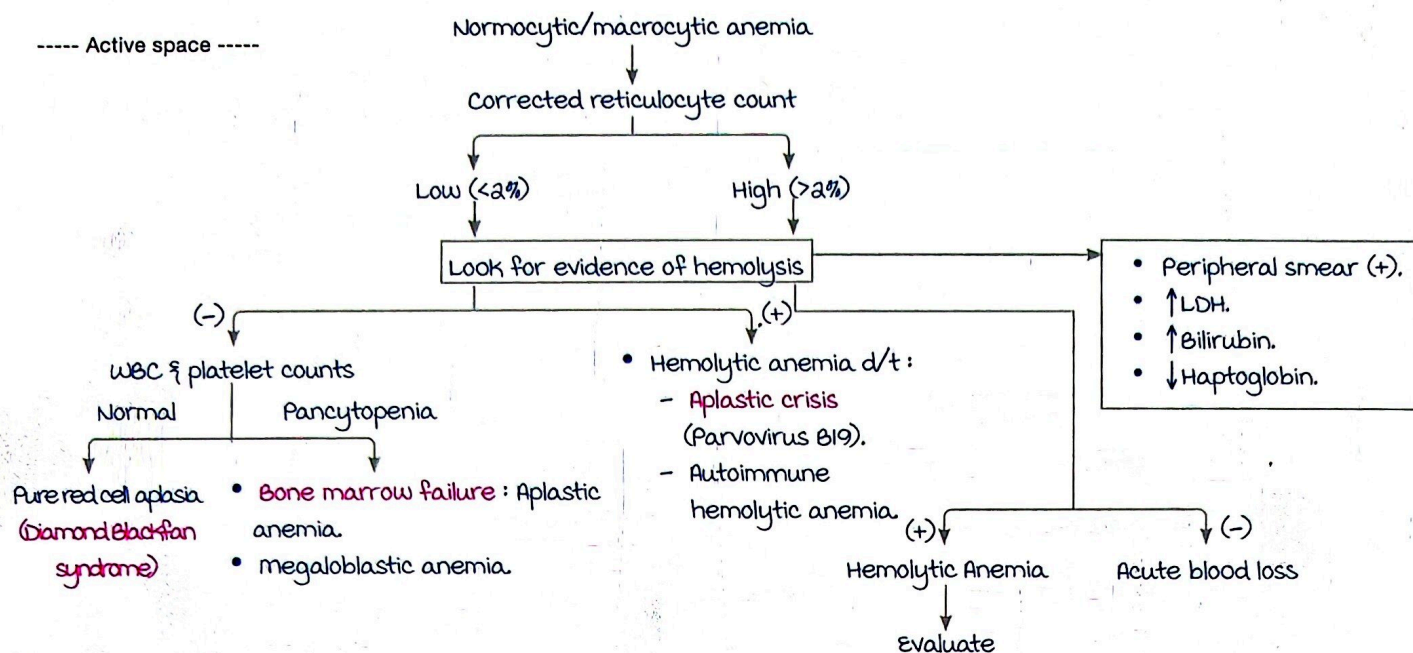


Normal/High mcv Anemia :

- Further evaluation based on **reticulocyte count** (Normal : 1-2%) → indicates bone marrow response.
- Corrected reticulocyte count = $\frac{\text{Reticulocyte count} \times \text{Patient's Hb}}{\text{Normal Hb for age/gender (Taken as 15)}}$ for degree of anemia.

>2% indicates adequate bone marrow response.

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Iron Deficiency Anemia

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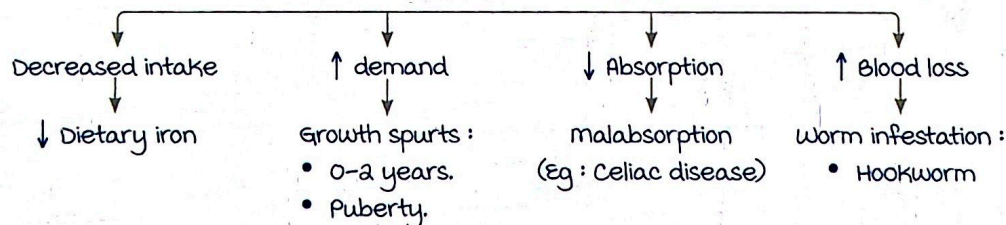
m/c cause of anemia in children worldwide.

PATHOGENESIS:

↓ Total body iron stores (↓ serum ferritin) → ↓ erythropoiesis → Anemia

Note: Iron stores ↓ only after a few months of ↓ dietary intake.

PREDISPOSING FACTORS:



Onset: 9-24 months.

- Late d/t good iron stores acquired from the mother in utero.

Rare causes of IDA:

- Atransferrinemia (Iron transfer to red cell precursors in bone marrow).
- Transferrin receptor antibodies.

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CLINICAL FEATURES :

- Tiredness.
- Easy fatigability.
- **most reliable** sign : Pallor.
 - Oral mucosa.
 - Nail beds.
 - Palpebral conjunctiva.
- Koilonychia : Spooning of nails.
- Platynychia : Flat nails.
- Atrophic glossitis : Shiny, smooth tongue.



Koilonychia

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Associations :

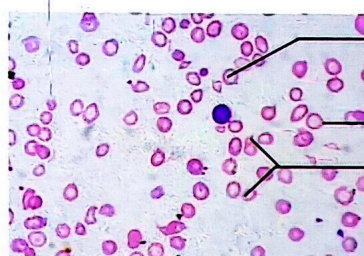
- Pica (Eating inedible substances).
- Temper tantrums.
- **Breath holding spells.**
- Restless leg syndrome.

INVESTIGATIONS :

1. Peripheral Smear :

2. Blood Parameters :

- Low MCV.
- **Low serum ferritin** (Screening).



Peripheral Smear findings : IDA

Hypochromia
(Pallor >1/3 of cell)

microcyte

Anisopoikilocytosis :
varying size & shape.

TREATMENT :

Ferrous Sulphate :

- most effective & economical form of iron.
- **20% elemental iron** present.
- Dose : 3-6 mg/kg/day (maximum: 200 mg/day).
- Duration : 4-6 months (To replenish iron stores).

Response to Iron Therapy :

Time after Iron Administration	Response
12-24 hours	Replenishment of intracellular iron enzymes. Clinically seen as subjective improvement .
36-48 hours	Bone marrow response : Erythroid hyperplasia
48-72 hours	Reticulocytosis : Peaks by 5-7 days
4-30 days	↑ Hb levels
1-3 months	Replenishment of iron stores

Supplementary Therapy :

- Dietary modification : Consumption of food rich in iron.
- Deworming : Albendazole.

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Megaloblastic Anemia

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macrocytes (Large RBCs) are seen on peripheral smear.

CAUSES :

Predisposing

↓ Vitamin B₁₂

↓ Seen in

- Pure vegetarian/vegan diet.
- Pernicious anemia (Auto-antibodies against intrinsic factor).
- malabsorption (Vit. B₁₂ : terminal ileum).
- **Inborn errors of metabolism :**
 - Homocystinuria.
 - Orotic aciduria.
 - methyl malonic acidemia.

↓ Folic acid (FA)

↓ Seen in

- Diet : ↑↑ Goat milk.
- Infection : **Giardiasis**.
- Drugs : Interfere with utilization of FA.
 - 6 mercaptopurine.
 - methotrexate.

CLINICAL FEATURES :

Pallor, tiredness & easy fatigability.

Characteristic Features :

- **Beefy red tongue.**
- Hepatosplenomegaly (30-40% cases).
- Knuckle hyperpigmentation.

Neurological manifestations :Seen in **Vit. B₁₂ deficiency** → ↓ methylation reactions in nervous system.Can occur **before onset of anemia**.**Features :**

- Loss of position/vibration sense : **earliest**.
- Ataxia : unsteadiness of gait.
- Paresthesia.
- Peripheral neuropathy.
- Subacute combined degeneration (SACD) of spinal cord.

} Characteristic

INVESTIGATIONS :**I. Peripheral Smear :**

- macro-ovalocytes : Large, oval RBCs.
- **Hypersegmented neutrophils** : At least 1 neutrophil ≥ 6 lobes on 5% neutrophils with 5 lobes.
 - D/t nuclear maturation delay.

Approach to Anemia in children & Nutritional Anemia

239

2. Complete Blood Count :

- Associated pancytopenia may be present.

3. Red Cell Indices :

- $\uparrow$ mcv (100-120 fL).



Peripheral Smear Findings

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TREATMENT :

For vit. B₁₂ deficiency :

- Vit. B₁₂ 1000 mcg i.m daily for first 2 weeks.



1000 mcg weekly until hematocrit normalizes.

- For pernicious anemia, monthly doses of vit. B₁₂ lifelong.
- Note : Folic acid without vit. B₁₂ supplementation $\rightarrow$ Persistence/worsening of neurological manifestations.
- In a case of megaloblastic anemia, treat with both vit. B₁₂ & folic acid.

For folic acid deficiency :

- Oral folic acid 1-5 mg daily x 3-4 weeks.
- Dietary supplementation.